



CUSTOMER SUPPORT

Submitted By

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MASTER OF SCIENCE

IN

INFORMATION TECHNOLOGY

Under the guidance

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May 2021



CERTIFICATE

This is to certify that the dissertation work entitled “**Customer Support**” is a record of bonafide research carried out by **Farhana A Rehim**, under my supervision and guidance as part of the partial fulfilment of the University regulation for the course of **M.Sc. in Information Technology**.

Date: June-2021

Place: Bengaluru

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DECLARATION

I, Farhana A Rehim, hereby declare that the dissertation report entitled “Customer Support” is a bonafide and genuine research work carried out by me under the guidance of Dr. Bosco Paul Alapatt. This report in part or full has not been submitted to this or any other University/College for the award of any degree or diploma.

Date: June 2021

Place: Bengaluru

Signature of Candidate

Farhana A Rehim

19MSCKR078

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FARHANA A REHIM

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ABSTRACT

The main aim of this project "Customer Support" is to help the employees in the call centre to handle their day to day calls. The app helps them to store the details of their customers, what product are they referring to and other details related to the product. There are specific module to list the customers who need follow up and another module for customers who has not answered the call.

The app display the full call history of that specific employee .Necessary changes to the data of the customer can be done based on calls. The employees have modules to display their details and they also have module to view their performance graph. There are modules to ease communication between manager and the employees.

The app is mainly focusing on reducing the burden of the employees by avoiding the confusion and giving them a clear picture on what all needs to be done for that particular day.

CHAPTER 1

1. INTRODUCTION

Call centres has become a very important part of our lives whether we like them or not. There are many different terms used to describe call centres - service centres, sales centres, contact centres etc. In its basic form the key role of a call centre is its responsibility for handling the telephone calls for an organization. For centres with more than 100 agents this may mean they handle several thousand calls per day. The calls may be inbound from customers or outbound made by the company to customers. This app is mainly to help the employees in the customer care centre to handle the calls. The individual can keep a record of the calls he made, can store the details about the call also in the app. They can divide the calls based on different criteria like those customers who has not picked up, those who need follow up etc. The app has a module to give a notification for employees on who all needs to be called today, the manager can send the daily target message to all employees and he can add new employees and view customer call list .Employees has a module to view their details and has a bar graph to view their performance in the past months.

The app is very user friendly, the employee just needs to login with their phone number and password .The individual can add details of customer he/she called the specific day, update details of customer based on calls made. The employee has different modules to check which all customer needs follow up, who all didn't answer calls etc.

1.1 OVERVIEW

“Customer support” is all about building an application that would help the employees in a call center to handle their customer calls efficiently. The reason why I chose this topic is that there are numerous app that helps to ease the communication between customer and employee but there are no apps that helps the employee to segregate and organize these calls. The app helps the employee to separate calls to different modules like not answered, callas that need follow up etc. There is a module to provide a notification on who all needs to be called today. The manager or head of the organization can send target messages to every employee through the app, he can add new employee and view customer list and there are modules to view the details and performance graph of employees.

The goal is to introduce an app for the employee for better customer as well as employee satisfaction in the organization. An organization can only expect maximum output from an employee if he/she is fully satisfied and happy and that is what we are trying to achieve here.

CHAPTER 2

2. SYSTEM STUDY AND ANALYSIS

2.1 PROBLEM STATEMENT

All the system that are currently available for the customer center concentrates on easing the communication between customer and employees. The employees at least have to attend 100 calls per day, report the customer issues and they also have to keep track on customers. There are no such app that helps the employees to ease such workload.

2.2 EXISTING SYSTEM

There are no existing system that provides the fragmentation of customer calls. The existing system eases the communication with customers but there are no module to fragment the data of customers.

2.2.1 DRAWBACKS

- I. Existing system does not have fragmentation of data.
- II. It does not provide separate modules to handle different type of calls.
- III. It does not contain modules that eases the communication between manager and employees

2.3 PROPOSED SYSTEM

The proposed system handles all the issues faced by existing system. The system provides fragmentation of data and reduces the workload for employees in an organization.

2.2.2 ADVANTAGE

- I. The system provides separate modules for each type of calls like follow up, not answered etc.
- II. The app contains module to ease communication between manager and employee.
- III. App contains modules to view employee details and their performance graph.
- IV. Each employee will be given their own dashboard.

2.4 OBJECTIVES OF THE PROJECT

- I. To propose a mobile application which helps the employee in a customer care center to manage their calls.
- II. To design and implement a mobile application that helps employee to add new customer details, track each customer and update their details based on calls.
- III. To implement an app that helps manager to add, delete and update employee, customers and send a broadcast message to all the employees.
- IV. To develop an application that helps employee view their details and performance graph.
- V. Several performance measures are used to investigate the result in order to validate the outcome of the project.

2.5 FEASIBILITY ANALYSIS

2.5.1 TECHNICAL FEASIBILITY

A study of function, performance and constraints may improve the ability to create an acceptable system. Technical feasibility is frequently the most difficult area to achieve at the stage of project engineering process. Considering that are normally associated with the technical feasibility include,

- Development risk
- Resource availability
- Technology

Technical feasibility study deals with the hardware as well as software requirements. This scope was whether the work for the project is done with the current requirements and existing software technology has to be examined in the feasibility study. The outcome was found to be positive. In the proposed system, data can be easily stored and managed using database management system software. The reports and results for various queries can be generated easily. Therefore, the system is technically feasible.

2.5.2 ECONOMIC FEASIBILITY

Proposed system was developed with the available resources. Since cost input for the application is almost nil the output of the application is always a profit. Hence software is

economically feasible. In the existing system, manpower is more required. In the proposed system, number of employees to be involved is reduced drastically. So, the proposed system is said to be economic. In the existing system, storage of the records should be properly done and security should be provided for the records. In the proposed system, the software provides security and maintenance and it hardly needs one or two persons to operate the system.

2.5.3 OPERATIONAL FEASIBILITY

Proposed projects are beneficial only if they can be turned into information system that will meet the organization's operating requirements. Simply stated, this test of feasibility asks if the system will work when it is developed and installed. The purpose of the operational feasibility study is to determine whether the new system will be used if it is developed and implementation from users that will undermine the possible application benefits.

There was no difficulty in implementing proposed system. It is so effective, user friendly and functionally reliable so that the users in the company will find the new system reduce their hard steps. If the user of the system is fully aware of the internal working of the system, then the users will not be facing any problem in running the system.

2.5.4 COST ESTIMATION AND SCHEDULING

The cost and schedule estimation process helps in determining number of resources to complete all project activities. It generally involves approximation and development of costing alternatives to plan, perform or work, deliver, or give project. A good estimation is very much essential for keeping a project under budget.

Two perspectives are generally required to derive project plans. These perspectives are given below:

Forward-Looking:

The Forward-Looking approach is also known as Top-Down approach.

This approach generally starts with describing and explaining various project tasks that involve starting with project aim or end deliverable and breaking it all down into smaller planning chunks.

Backward-Looking:

Backward-Looking approach is also known as Bottom-up approach.

In this approach, project team breaks requirements of clients down, determining lowest level appropriate to develop a range of estimates, covering overall scope of project based on available definition of task.

2.6 SCOPE AND SIGNIFICANCE OF THE PROJECT

Though numerous methods have been proposed to manage calls in a center, the feature that makes us stand out from others is fragmentation of the data.

In this mobile application we have assigned separate modules for each type of calls, which makes it easier for the employee to select the specific list of customers. That is if the employee only wants the detail of customers who hasn't answered the phone then we select the specific module rather than searching the whole customer list. Admin/Manager can easily communicate with all the employees. The employee has a performance graph to view or compare their performance in months. These modules makes it easier for them to handle their work easily. This motivated me to develop "Customer Support" which could simplify the work load for the employees in an organization.

2.7 LITERATURE REVIEW

- Lawrence and Francis (2005) discussed the need for a proper CRM software in organizations and their importance in the growth of the company.
- George and Clive (2006) explored call centre management from the perspective of the managers, particularly what the key management responsibilities are in managing call centres and the key performance indicators (KPIs) used in managing call centres.
- Zeynep, et al. (2007) explored about the operational challenges faced by call center managers. Issues associated with human resources management, sales, and marketing have also become increasingly relevant to call center operations and associated academic research. In this paper, they provide a survey of the recent literature on call center operations management.
- Frantz, et al. (2011) explored on a quantitative survey of 155 call center managers in France, which benefited from extensive preparation through ten case studies.

- Sartika (2015) designed a web based customer service information system to improve service quality and efficiency in a call center.
- Mary (2016) explored issues related to the design of information systems to support customer feedback from an organizational information processing perspective.
- Shiva (2017) developed a call center management system to register calls, make announcements. Their main form is an MDI application.
- Morteza, et al. (2017) studied about the the current literature on contact centers and identify their shortcomings to be addressed in the current digital age.
- Divya (2018) explored issues related to the design of information systems to support customer feedback from an organizational information processing perspective.
- Wei, et al. (2018) analyzed the three major problems of call center service (cask effect, avalanche effect, cumulative effect), introduces the call center platform performance testing strategy, and uses the Sprint and SilkPerformer testing tools to test the call center platform.
- Sheethal (2019) introduced an automated system that helps in routing calls and data collection.
- Hermenegildo (2019) explored the understanding of customer relationship management (CRM) as a set of technological solutions key for efficient business management.

CHAPTER 3

3. DEVELOPMENT ENVIRONMENT

3.1 HARDWARE REQUIREMENT

HARDWARE:

CPU type	: Intel Core i5
RAM size	: 512 MB
HARD DISK capacity	: 40 GB
Monitor Type	: 15 inch color monitor
Mobile	: Android Phone
Keyboard	: Internet Keyboard

3.2 SOFTWARE REQUIREMENT

SOFTWARE:

1. Android Studio(fron end)
2. PHP - Sublime Editor
3. MySQL (back end)
4. Operating System : Windows XP / Windows7/ Windows Vista
5. Web Browser: Microsoft Internet Explorer, Mozilla, Google Chrome or later
6. HTML

CHAPTER 4

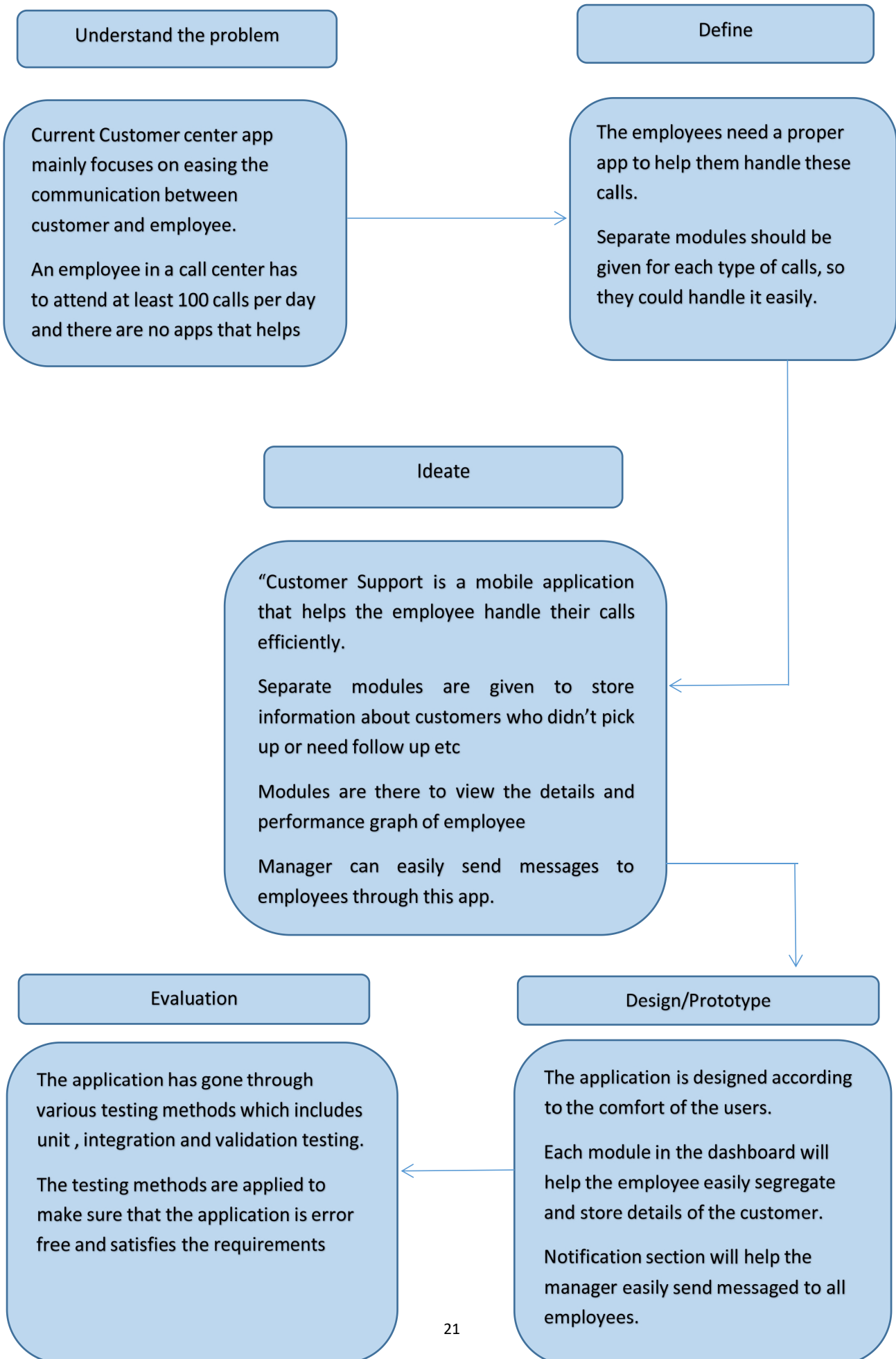
4. SYSTEM DESIGN

4.1 ELEMENTS OF DESIGN

Design elements are the basic units of any visual design which form its structure and convey visual messages. Elements of design defined as line, direction, shape size, texture, value and colour, concluding that these elements are the material from which all designs are built.

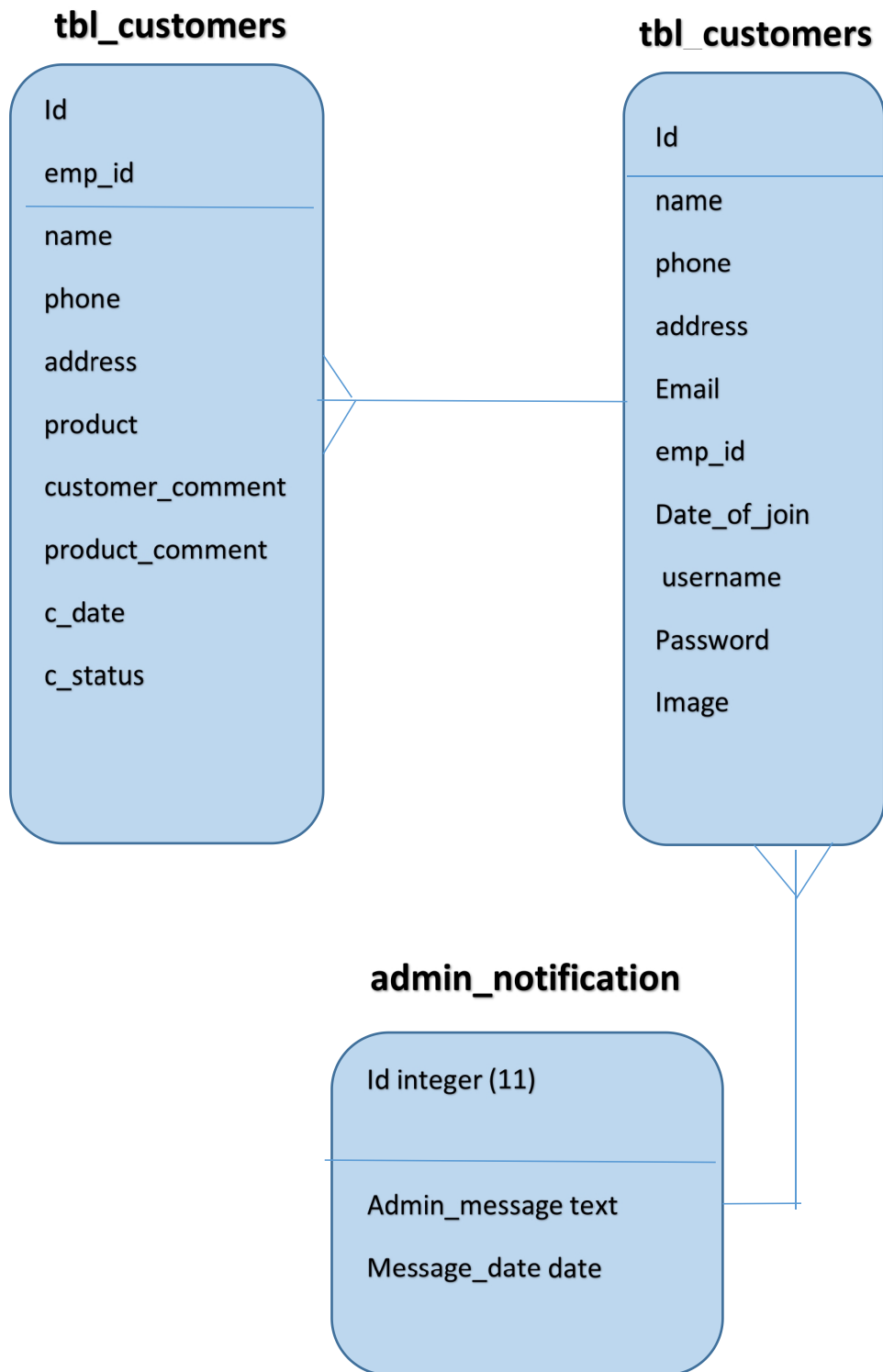
4.1.1 PROCESS DESIGN

Process design is the act of transforming an organization's vision, goals and available resources into a discernible, measurable means of achieving the organization's vision.



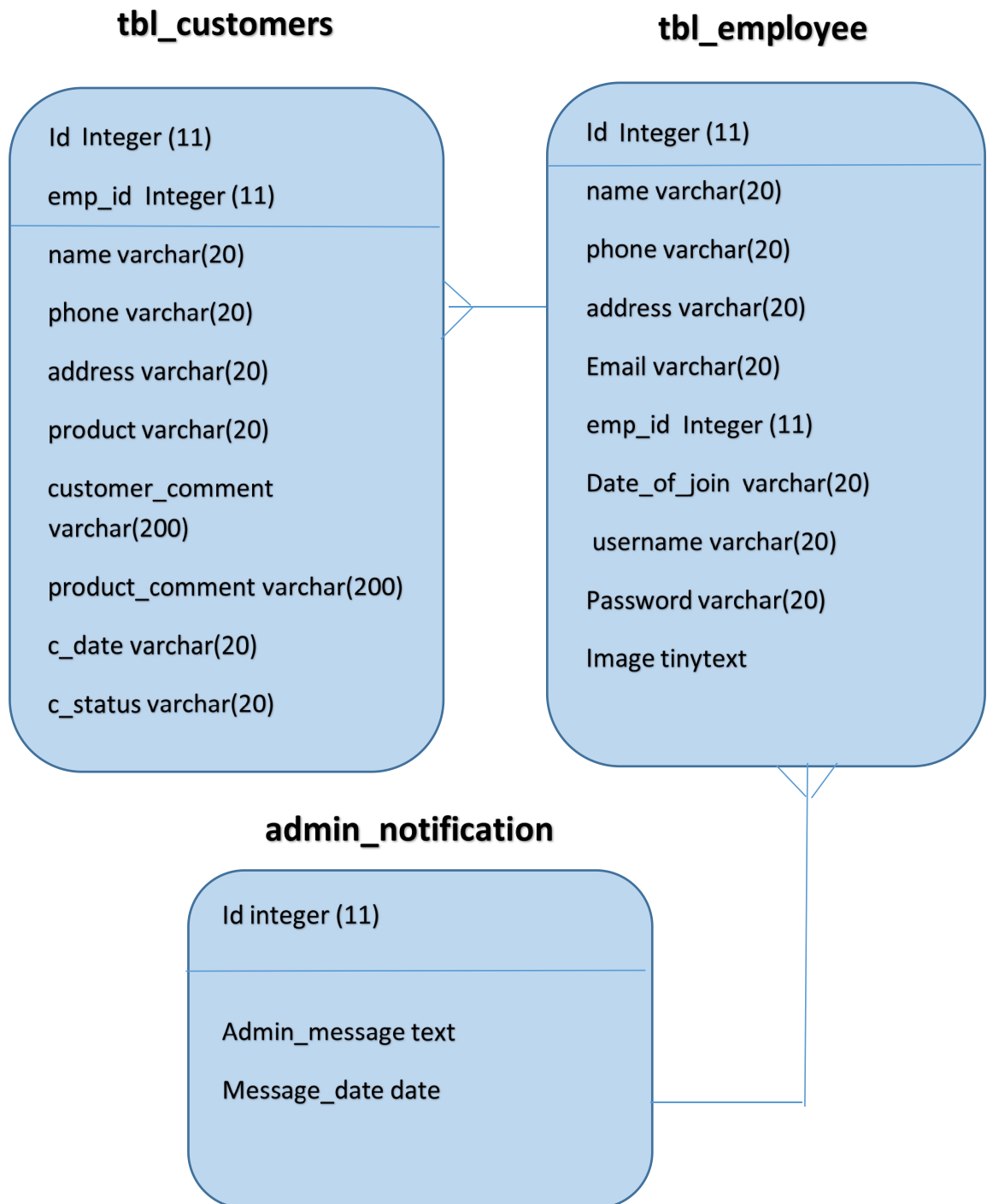
4.1.2 LOGICAL DESIGN

A logical design is a conceptual abstract design. The process of logical design involves arranging data into a series of logical relationships called entities and attributes.



4.1.3 PHYSICAL DESIGN

Physical design is the process of determining the data storage organization and data access characteristics of the database in order to ensure its integrity, security, and performance. This is the last stage in the database design process.



4.1.4 INPUT DESIGN

In this application we are only taking employee username, password and customer details as input. Once the employee has logged in with his/her credentials they can select “ add customer” module to add details of their customers. There is an option to update these details in “ not answered” and “follow up” module. The admin can add new employee or customers through back end as well.

11:39 89%

< Update Details

Name :
ajay

Phone :
9447508721

Address :
test

Product :
fridge

Customer Comment :
test

Product Comment :
test

Date :
2021-05-05

Status :
Follow Up

SUBMIT

11:38 89%

< Add Customer

Name :

Phone :

Address :

Product :

Customer Comment :

Employee Comment :

Date :
2021-05-05

Status :
Completed

SUBMIT

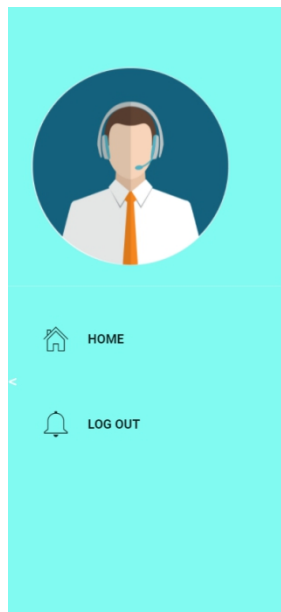
11:39 89%

Customer Support

Username

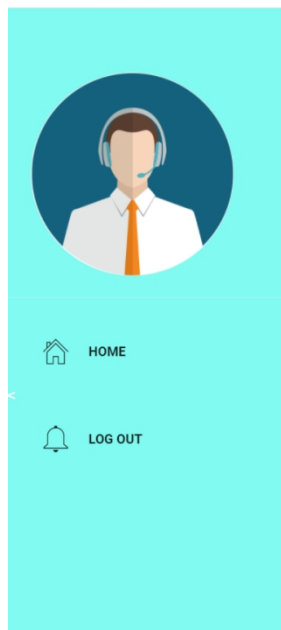
Password

Login



Customers Details

ID		NAME	
PHONE		ADDRESS	
PRODUCT		CUSTOMER COMMENT	
PRODUCT COMMENT		C DATE	18-05-2021
C STATUS	follow up	EMP ID	

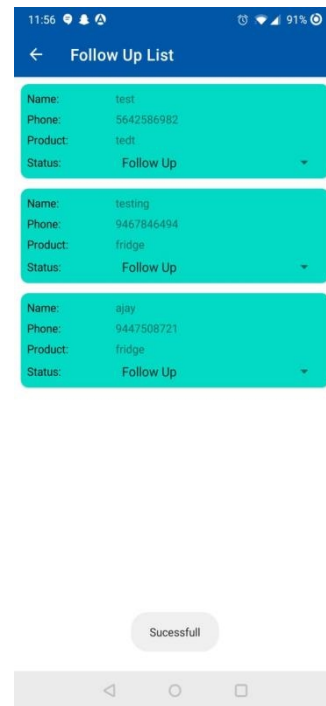


Employee Details

NAME		PHONE	
ADDRESS		EMAIL	
DATE OF JOIN		EMP ID	
USERNAME		PASSWORD	
IMAGE	Choose file No file chosen		

1.5 OUTPUT DESIGN

“Customer Support” display the details of all the customers the specific employee has called as a list. There are separate modules to display customers who has not answered and those who needs follow up. Employee performance graph and details are displayed in modules.



localhost/CustomerCare/admin/tbl_customers/select.php

Apps New YouTube Gmail Maps Reading list

SHOW 10 ENTRIES SEARCH:

Add New

ID	NAME	PHONE	ADDRESS	PRODUCT	CUSTOMER COMMENT	PRODUCT COMMENT	C DATE	C STATUS	EMP ID	UPDATE	DEL
1	test	5642586982	test	tedt	tedt	yedtte	2021-04-30	Completed	1	Update	Del
4	ii	666666	hhv bhi	CNN go	CNN	chhi	2021-05-26	Completed	1	Update	Del
5	testing	9467846494	testing	fridge	testing	testing	2021-05-03	Follow Up	1	Update	Del
6	t	9457846134	t	t	t	t	2021-05-05	Follow Up	1	Update	Del
7	ajay	9447508721	test	fridge	test	test	2021-05-05	Follow Up	1	Update	Del
8	gopika	9457816138	test	fridge	test	test	2021-05-08	Follow Up	1	Update	Del
9	farhana	9447508721	test	washing machine	test	test	2021-05-12	Follow Up	1	Update	Del
10	amal T	9847039693	test	fridae	test	test	2021-	Not	1	Udate	Del

HOME LOG OUT

localhost/CustomerCare/admin/tbl_employee/select.php

Apps New YouTube Gmail Maps Reading list



[HOME](#)
[LOG OUT](#)

SHOW 10 ENTRIES

SEARCH:

Add New

ID	NAME	PHONE	ADDRESS	EMAIL	DATE OF JOIN	EMP ID	USERNAME	PASSWORD	IMAGE	UPC
1	test	9847039693	test	test1234@gmail.com	2-4-2019	e001	test	test	profile.png	Upc
2	farhana	9447508721	al harmont , t square road , pachalam	farhana.arehim@gmail.com	26-12-2019	2	farah	qwerty	test1.jpg	Upc
3	emp1	09447508721	al harmony, t square road , pachalam	farhana.arehim@gmail.com	12-04-2021	e003	emp	emp	2021-05-09-11-04c1.jpg	Upc
4	naseedha	8273738292	test	neenu.shamna@gmail.com	12-04-2021	e004	neenu	neenu	2021-05-12-00-45c1.jpg	Upc

Showing 1 to 4 of 4 entries

Previous 1 Next

4.1.6 DATABASE DESIGN

This application access three tables from the database. Tbl_customers for storing customer details, tbl_employee for storing employee details and admin_notifications for storing notification messaged from admin/manager.

Tbl_customers

	id	name	phone	address	product	customer_comment	product_comment	c_date	c_status	emp_id
☐ Edit Copy Delete	1	test	5642586982	test	tedt	tedt	yedtt	2021-04-30	Follow Up	1
☐ Edit Copy Delete	4	ii	666666	h hv bhi	CNN go	CNN	chhi	2021-05-26	Completed	1
☐ Edit Copy Delete	5	testing	9467846494	testing	fridge	testing	testing	2021-05-03	Follow Up	1
☐ Edit Copy Delete	6	t	9457846134	t	t	t	t	2021-05-05	Not Answered	1
☐ Edit Copy Delete	7	ajay	9447508721	test	fridge	test	test	2021-05-05	Follow Up	1

tbl_employee

 Edit  Copy  Delete	id	name	phone	address	email	date_of_join	emp_id	username	password	image
☐	1	test	9847039693	test	test123@gmail.com	2-4-2019	e001	test	test	profile.png
☐	2	farhana	9447508721	al harmont , t square road , pachalam	farhana.arehim@gmail.com	26-12-2019	2	farah	qwerty	test1.jpg
☐	3	emp1	09447508721	al harmony, t square road , pachalam	farhana.arehim@gmail.com	12-04-2021	e003	emp	emp	2021-05-05-09-11-04c1.jpg
☐	4	naseedha	8273738292	test	neenu.shamna@gmail.com	12-04-2021	e004	neenu	neenu	2021-05-05-12-00-45c1.jpg

admin_notifications

+ Options

 Edit  Copy  Delete	id	admin_message	message_date
☐	1	07-Jul-2012 — Select count(distinct(col1 col2))...	2021-04-26
☐	2	test message	2021-05-03
☐	3	test test	2021-05-03
☐	4	100 calls per employee	2021-05-05

CHAPTER 5

5. SYSTEM REQUIREMENT SPECIFICATION

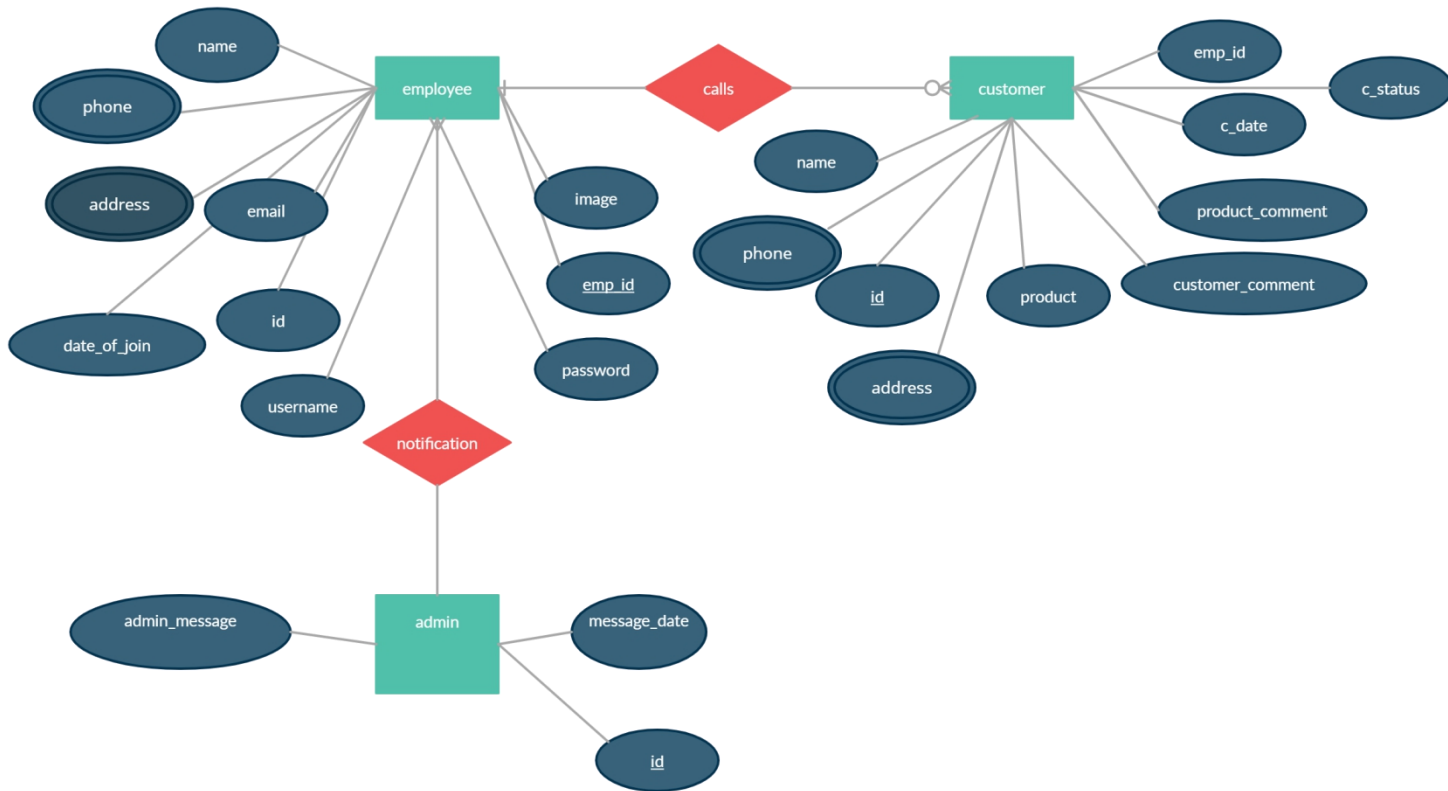
The purpose of this SRS is to specify the requirements of the mobile based application. This software requirements specification provides a complete description of all the functions and specification modules. The below document contains the software requirements of “Customer Support”.

5.1 SYSTEM DESIGN

Systems design is the process of defining elements of a system like modules, architecture, components and their interfaces and data for a system based on the specified requirements. It is the process of defining, developing and designing systems which satisfies the specific needs and requirements of a business or organization.

5.1.1 E R DIAGRAM

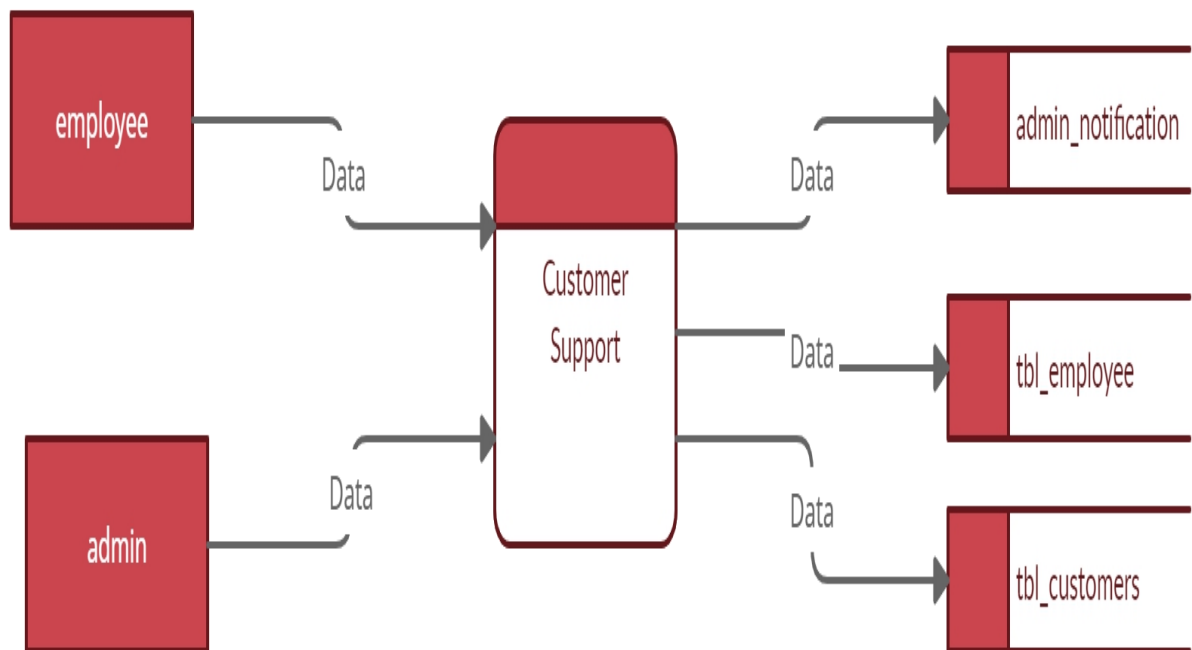
An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research.



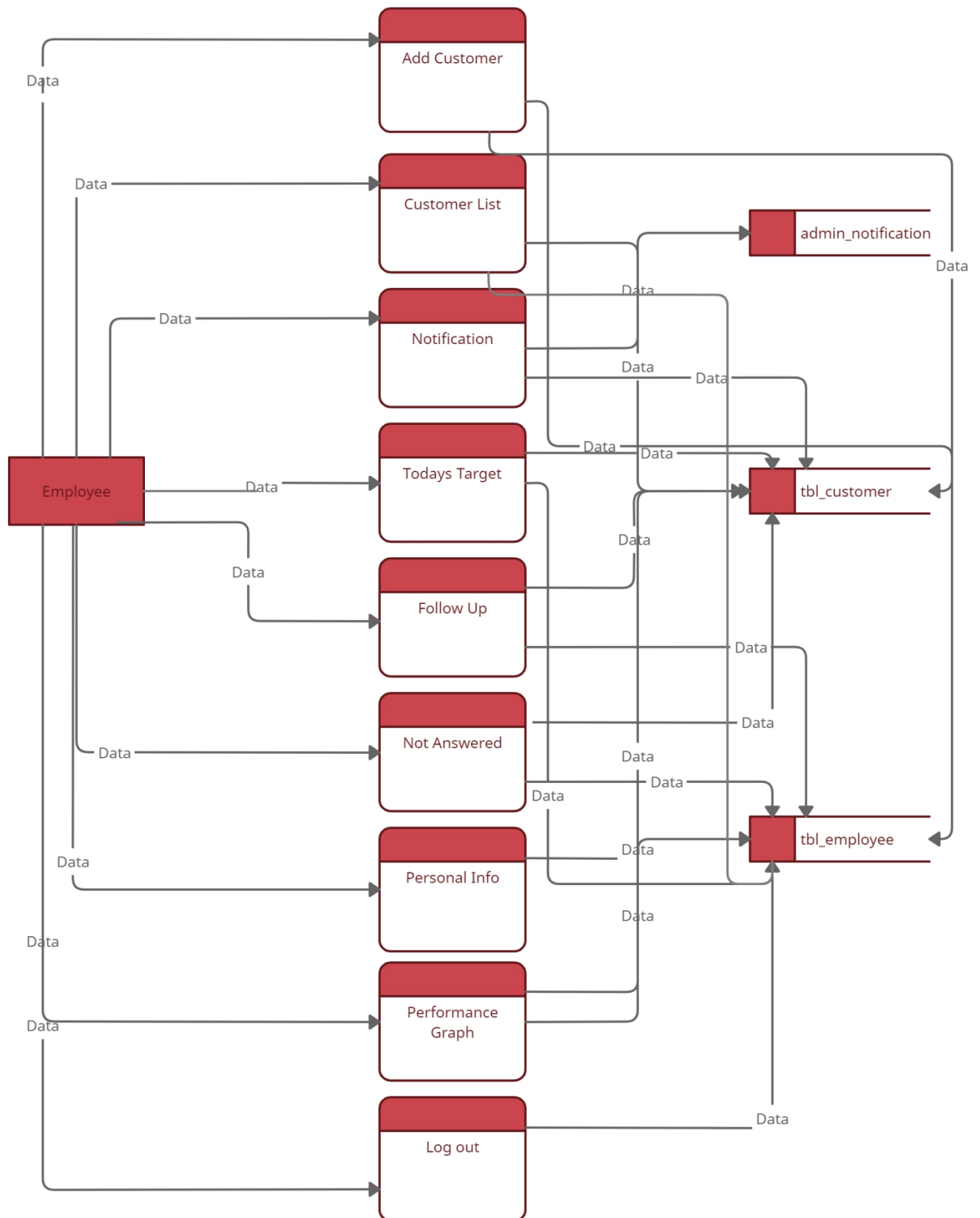
5.1.2 DATA FLOW DIAGRAM

DFD graphically representing the functions, or processes, which capture, manipulate, store, and distribute data between a system and its environment and between components of a system. The visual representation makes it a good communication tool between User and System designer. Structure of DFD allows starting from a broad overview and expand it to a hierarchy of detailed diagrams.

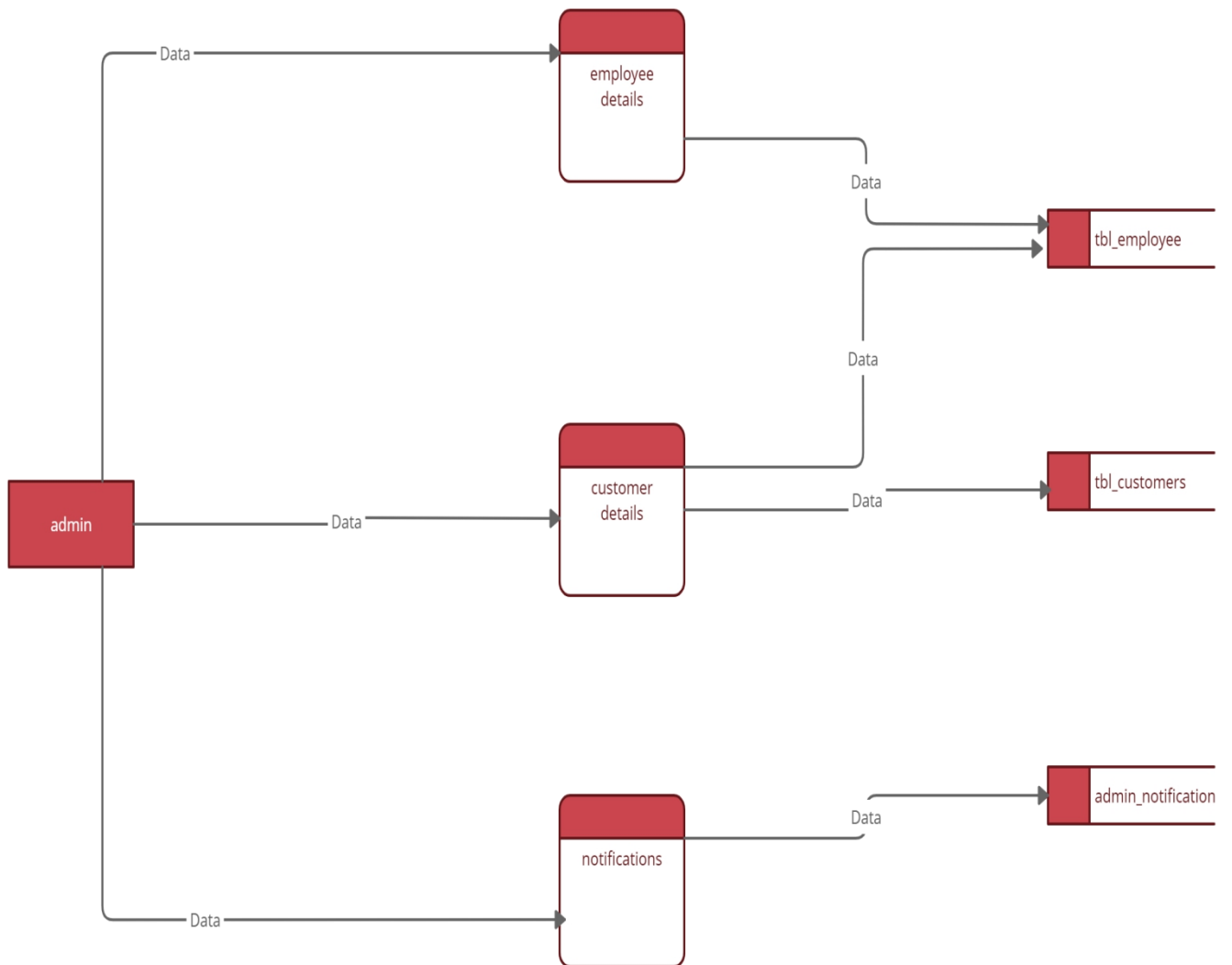
CONTEXT LEVEL



LEVEL 1 DFD

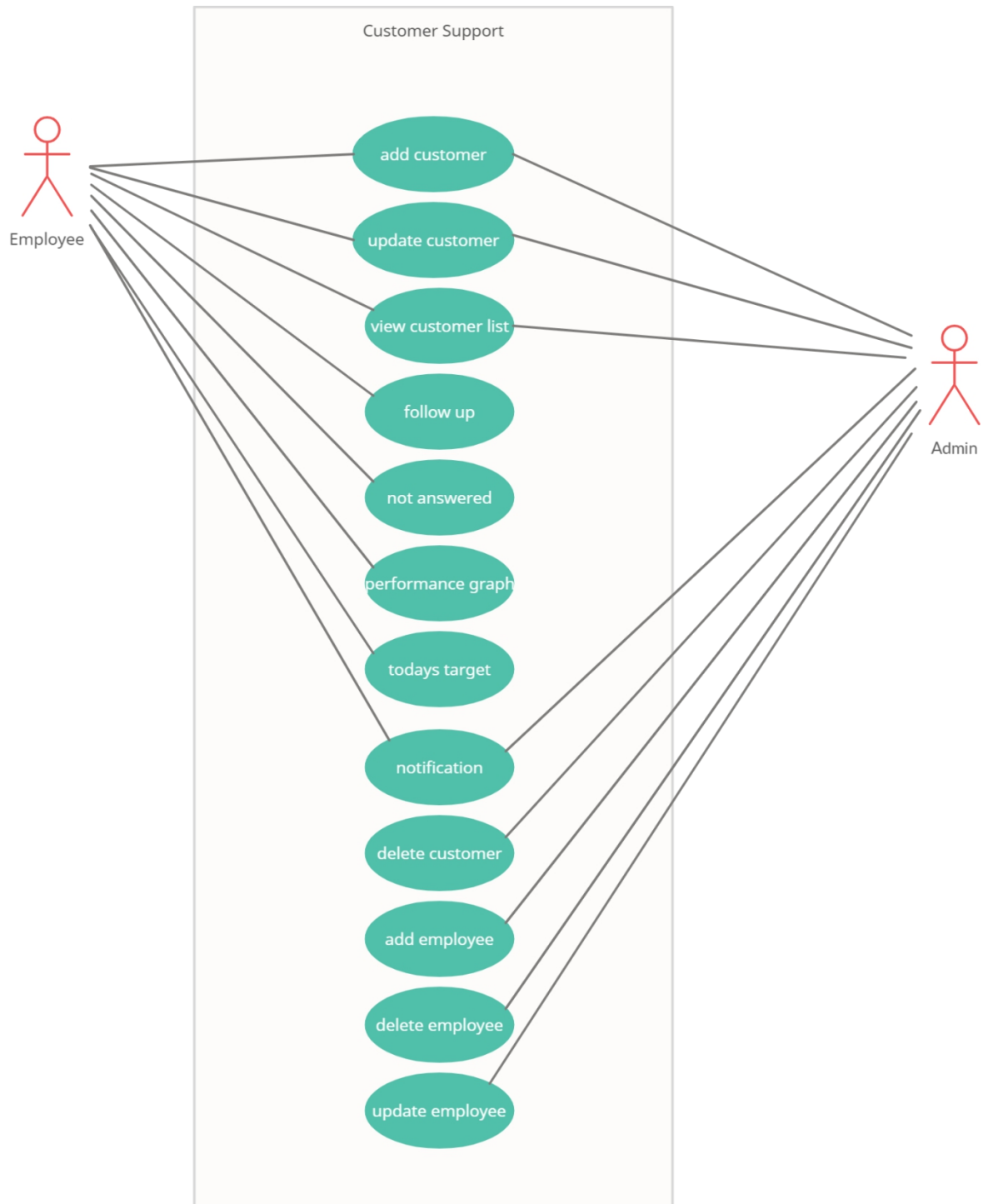


LEVEL 2 DFD



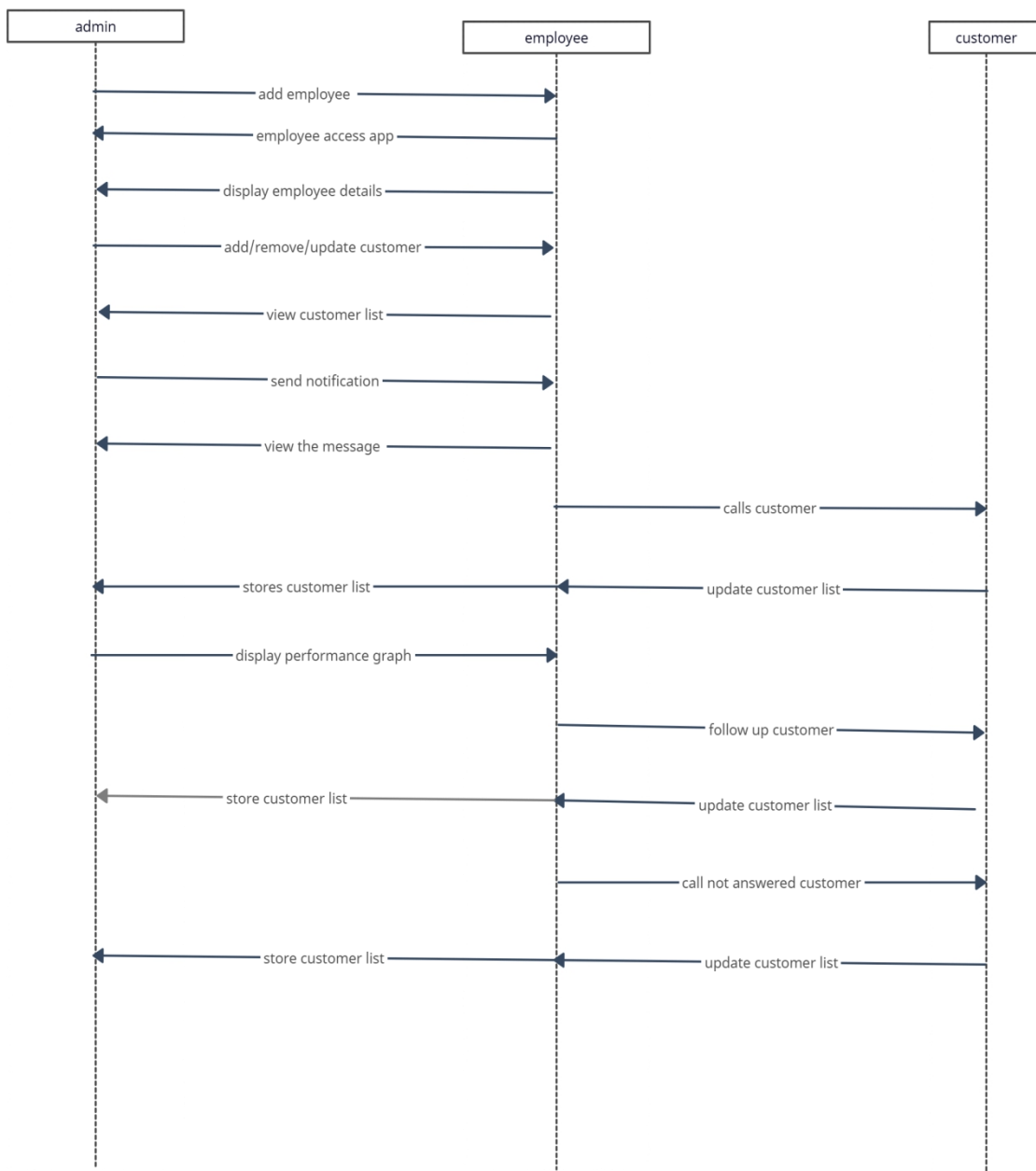
5.1.3 USECASE DIAGRAM

A use case diagram is a graphical depiction of a user's possible interactions with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. The actors are often shown as stick figures.



5.1.4 SEQUENCE DIAGRAM

A sequence diagram simply depicts interaction between objects in a sequential order i.e. the order in which these interactions take place. We can also use the terms event diagrams or event scenarios to refer to a sequence diagram. Sequence diagrams describe how and in what order the objects in a system function. These diagrams are widely used by businessmen and software developers to document and understand requirements for new and existing systems.



CHAPTER 6

6. SYSTEM TESTING AND IMPLEMENTATION

6.1 SYSTEM TESTING

It is a level of testing that validates the complete and fully integrated software product. The purpose of a system test is to evaluate the end to end system specifications. In this application all the 8 modules together were tested and approved.

6.1.1 UNIT TESTING

It is a type of software testing where individual unit or components of a software is tested. I have used this testing method to check the performance of each module individually. For example let us consider the module “login”, here I have tested if the data entered is invalid then will an error message be display and if it is valid credentials then will it login to the dashboard.

```
public void onResponse(String response) {

    Log.d(TAG, msg: "onResponse: "+response);

    if(response.contains("Failed")){
        Toast.makeText( context: LoginActivity.this, text: ""+response, Toast.LENGTH_SHORT).show();
    }
    else {

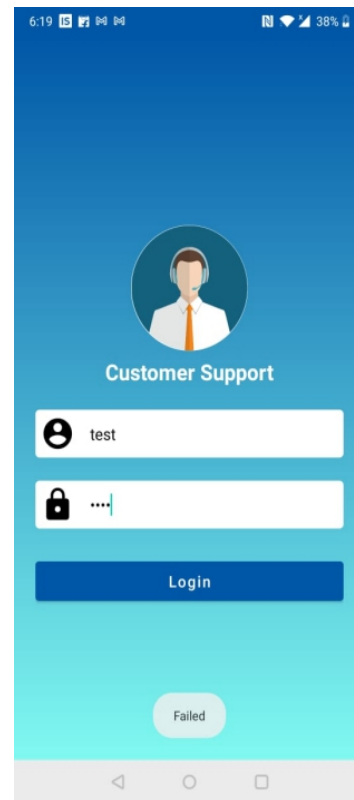
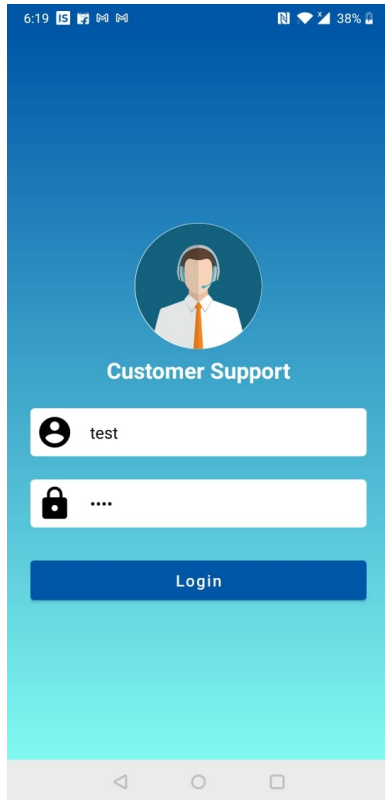
        String image = "";
        try {

            JSONObject jsonObject = new JSONObject(response);
            JSONArray jsonArray = jsonObject.getJSONArray( name: "data");

            for (int i = 0; i < jsonArray.length(); i++) {
                JSONObject object = jsonArray.getJSONObject(i);
                globalPreference.addUID(object.getString( name: "id"));
                globalPreference.addUname(object.getString( name: "name"));
                globalPreference.addUIImage(object.getString( name: "image"));
            }

            globalPreference.setLoginStatus(true);

        } catch (JSONException e) {
            e.printStackTrace();
        }
    }
}
```



6.1.2 INTEGRATION TESTING

It is a type of software testing in which individual software modules are combined and tested as a group. This method has been implemented in my project to check the functioning of two modules that is “Add customer” and “Customer List”.

The details of one customer is taken stored in the database using “ Add customer” module and after that checked if the new customer details were printed in the “Customer list” module.

```

@Override
protected Map<String, String> getParams() throws AuthFailureError {
    Map<String, String> params = new HashMap<>();
    params.put("customerName", customerNameET.getText().toString());
    params.put("customerPhone", customerPhoneET.getText().toString());
    params.put("customerAddress", customerAddressET.getText().toString());
    params.put("customerProduct", customerProductET.getText().toString());
    params.put("customerComment", customerCommentET.getText().toString());
    params.put("customerProductComment", customerProductCommentET.getText().toString());
    params.put("customerDate", customerDateTV.getText().toString());
    params.put("customerStatus", customerStatusSpinner.getSelectedItem().toString());
    params.put("empid", globalPreference.RetrieveUID());
    return params;
}

```

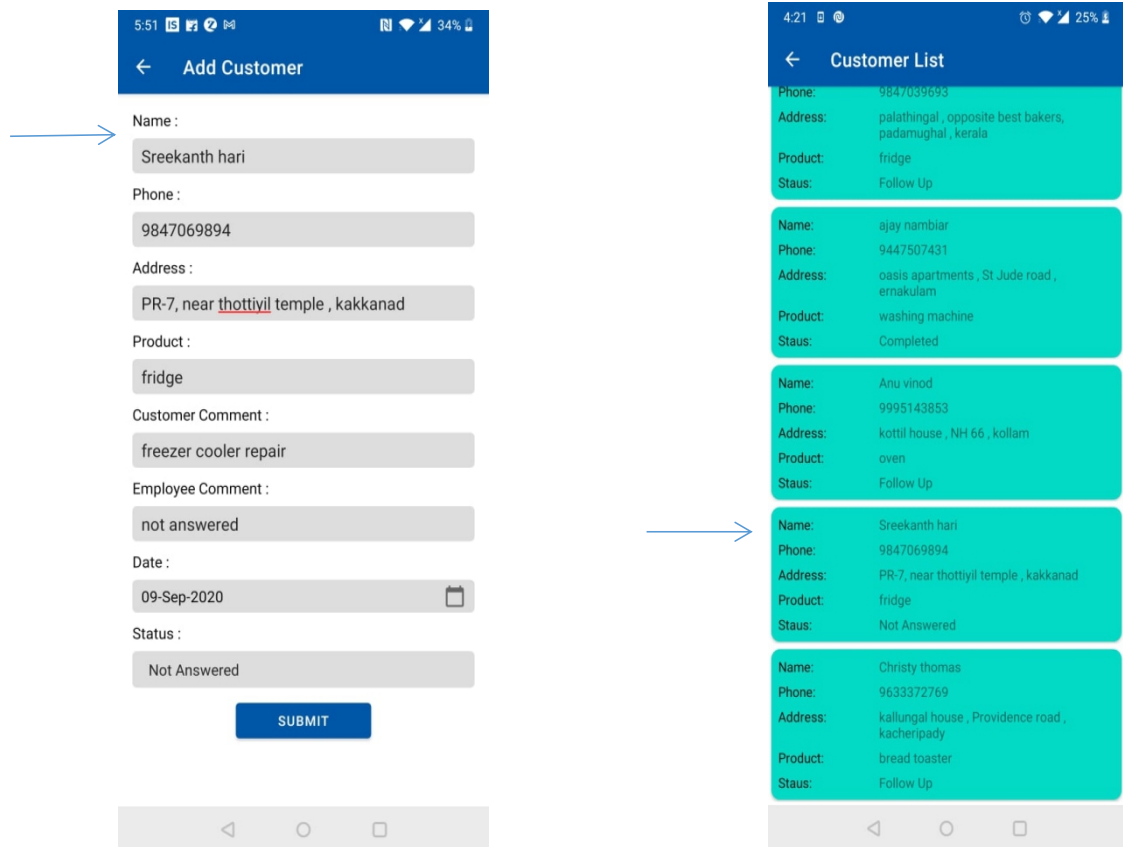
```

try {
    JSONObject jsonObject = new JSONObject(response);
    JSONArray jsonArray = jsonObject.getJSONArray( name: "data");
    for (int i = 0; i < jsonArray.length(); i++)
    {
        JSONObject object = jsonArray.getJSONObject(i);
        String id = object.getString( name: "id");
        String name = object.getString( name: "name");
        String phone = object.getString( name: "phone");
        String address = object.getString( name: "address");
        String product = object.getString( name: "product");
        String customer_comment = object.getString( name: "customer_comment");
        String product_comment = object.getString( name: "product_comment");
        String c_date = object.getString( name: "c_date");
        String c_status = object.getString( name: "c_status");

        customerDetailsList.add(new CustomerDetailsModelClass(id,name,phone,address,product,customer_comment,product_comment,c_date,c_status));
    }
} catch (JSONException e) {
    e.printStackTrace();
}

CustomerListAdapter adapter = new CustomerListAdapter(customerDetailsList, context: CustomerListActiv
customerListRV.setAdapter(adapter);

```



6.1.3 USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) is a type of testing performed by the end user or the client to verify/accept the software system before moving the software application to the production environment. UAT is done in the final phase of testing after functional, integration and system testing is done.

6.1.4 VALIDATION TESTING

The process of evaluating software during the development process or at the end of the development process to determine whether it satisfies specified business requirements.

In this mobile application I have validated the details of customers in the add customer module before it is stored in the database. For example only numbers up to 10 digit can be added in phone section. A calendar option is given at the date section so that the data would only be stored in date format. Most of the column cannot be left unfilled.

```

public void onClick(View v) {
    if(customerNameET.getText().toString().equals(""))
    {
        customerNameET.setError("Please Fill");
    }
    else if(customerPhoneET.getText().toString().equals(""))
    {
        customerPhoneET.setError("Please Fill");
    }
    else if(customerAddressET.getText().toString().equals(""))
    {
        customerAddressET.setError("Please Fill");
    }
    else if(customerProductET.getText().toString().equals(""))
    {
        customerProductET.setError("Please Fill");
    }
}/*

```

```

<EditText
    android:id="@+id/customerPhoneEditText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:paddingLeft="10dp"
    android:paddingRight="10dp"
    android:inputType="phone"
    android:maxLength="10"
    android:paddingTop="6dp"
    android:paddingBottom="6dp"
    android:background="@drawable/rect_grey"/>

```

<TextView

```
    android:id="@+id/customerDateTextView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="8dp"
    android:paddingLeft="10dp"
    android:paddingRight="10dp"
    android:layout_gravity="center"
    android:gravity="center|start"
    android:textColor="@color/black"
    android:textSize="16sp"
    android:paddingTop="5dp"
    android:paddingBottom="5dp"
    android:drawableRight="@drawable/ic_calendar"
    android:background="@drawable/rect_grey"/>
```

5:02 22%

← Add Customer

Name :

Phone :

Address :

Product :

Customer Comment :

Employee Comment :

Date :

Status :

SUBMIT

6.2 SYSTEM SECURITY

System security encompasses all facets of accessing information assets. From authentication, to software updates, anti-virus protection, and modifications - security is a key component to a device operating at its optimum. These best practices help to mitigate various security concerns.

In “Customer Support “it is made sure that employee can only access the app using their username and password.

```
public void onResponse(String response) {

    Log.d(TAG, "msg: " + "onResponse: " + response);

    if(response.contains("Failed")){
        Toast.makeText(context, LoginActivity.this, " " + response, Toast.LENGTH_SHORT).show();
    }
    else {

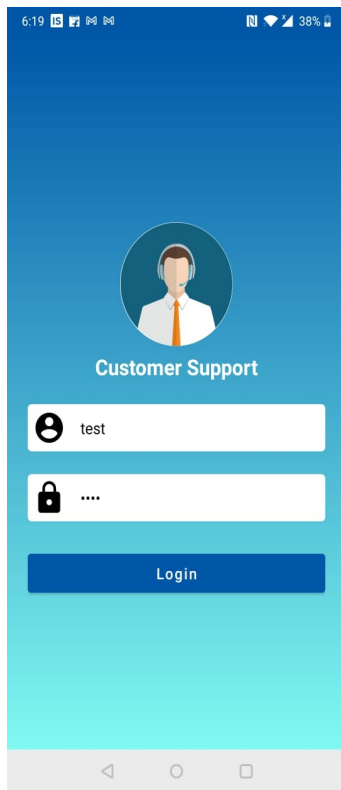
        String image = "";
        try {

            JSONObject jsonObject = new JSONObject(response);
            JSONArray jsonArray = jsonObject.getJSONArray( name: "data");

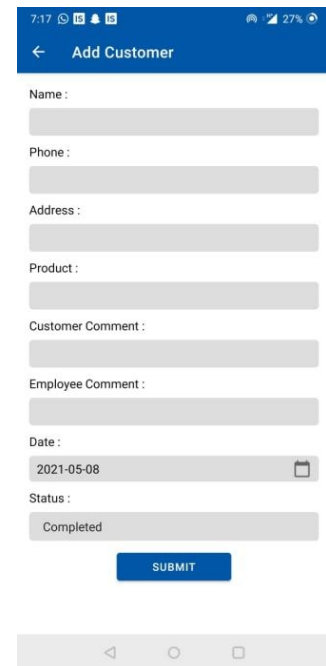
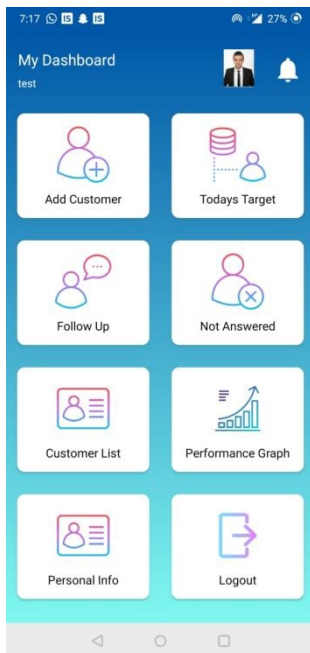
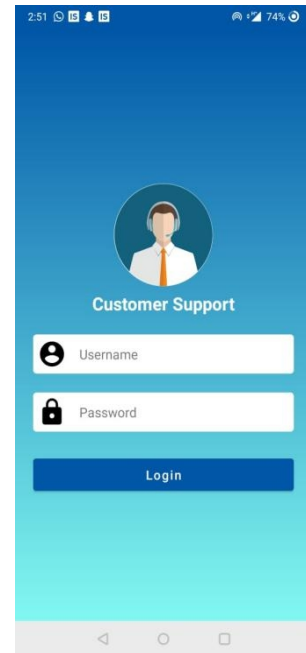
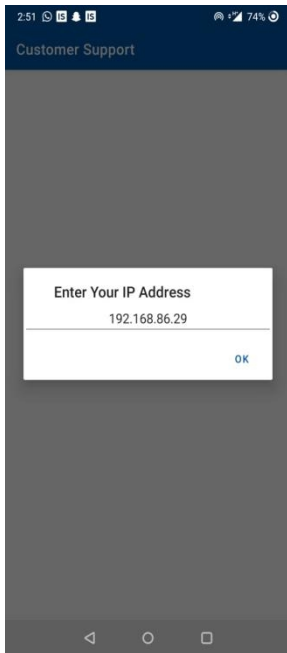
            for (int i = 0; i < jsonArray.length(); i++) {
                JSONObject object = jsonArray.getJSONObject(i);
                globalPreference.addUID(object.getString( name: "id"));
                globalPreference.addUname(object.getString( name: "name"));
                globalPreference.addUIImage(object.getString( name: "image"));
            }

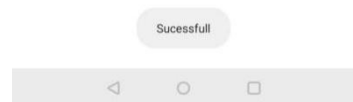
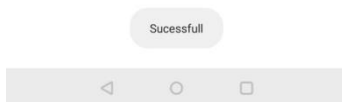
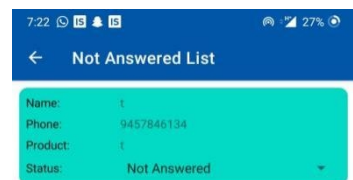
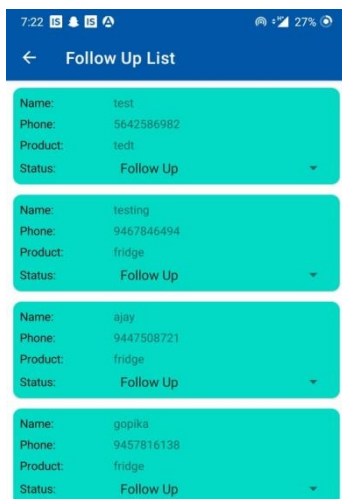
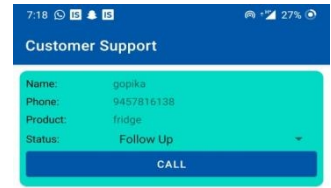
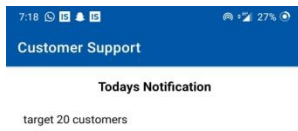
            globalPreference.setLoginStatus(true);

        } catch (JSONException e) {
            e.printStackTrace();
        }
    }
}
```



6.3 SNAPSHOTS



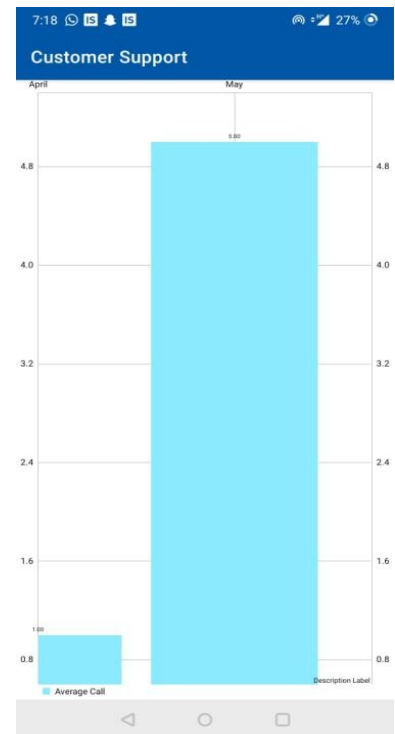


7:22
15
27%

Employee Profile



Name: test
Phone: 9847039693
Email: test123@gmail.com
Address: test
Date of join: 2-4-2019
Employee ID: e001

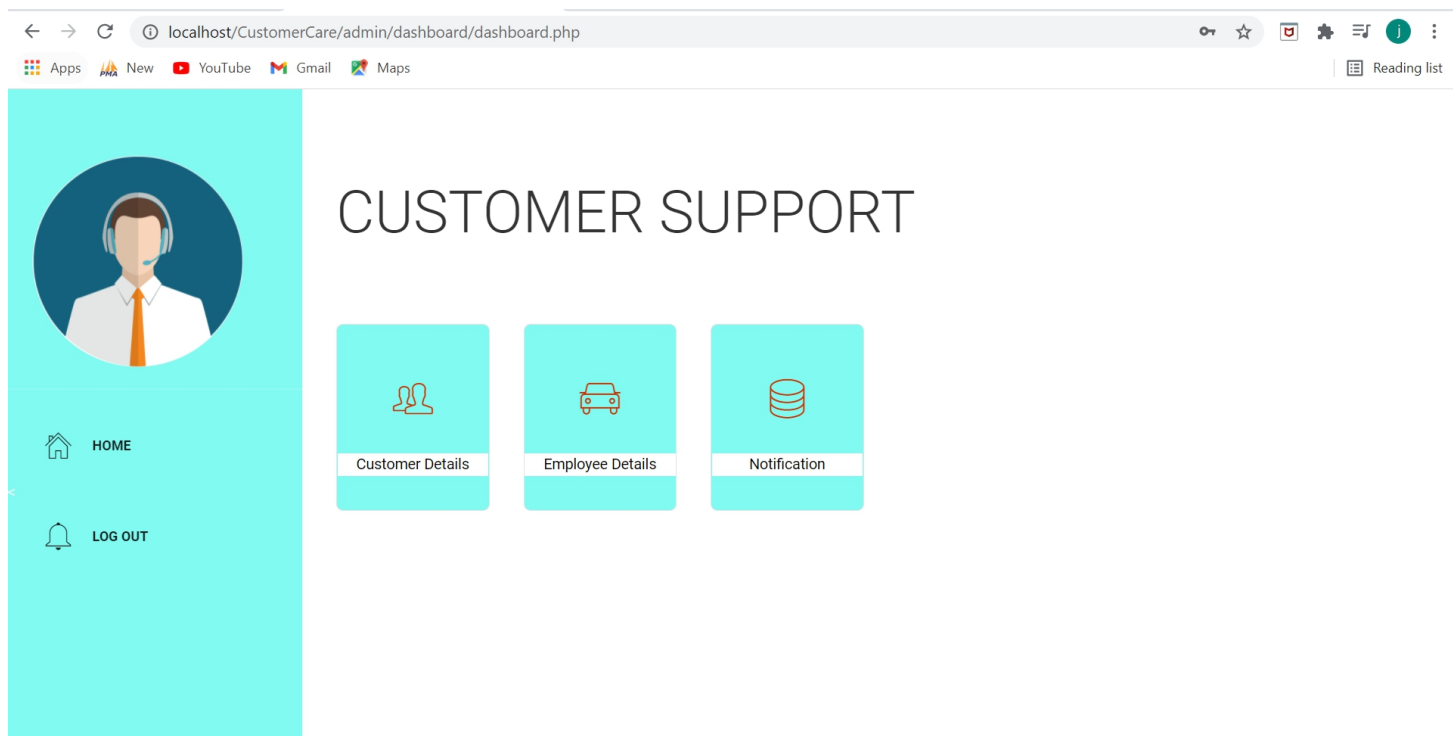
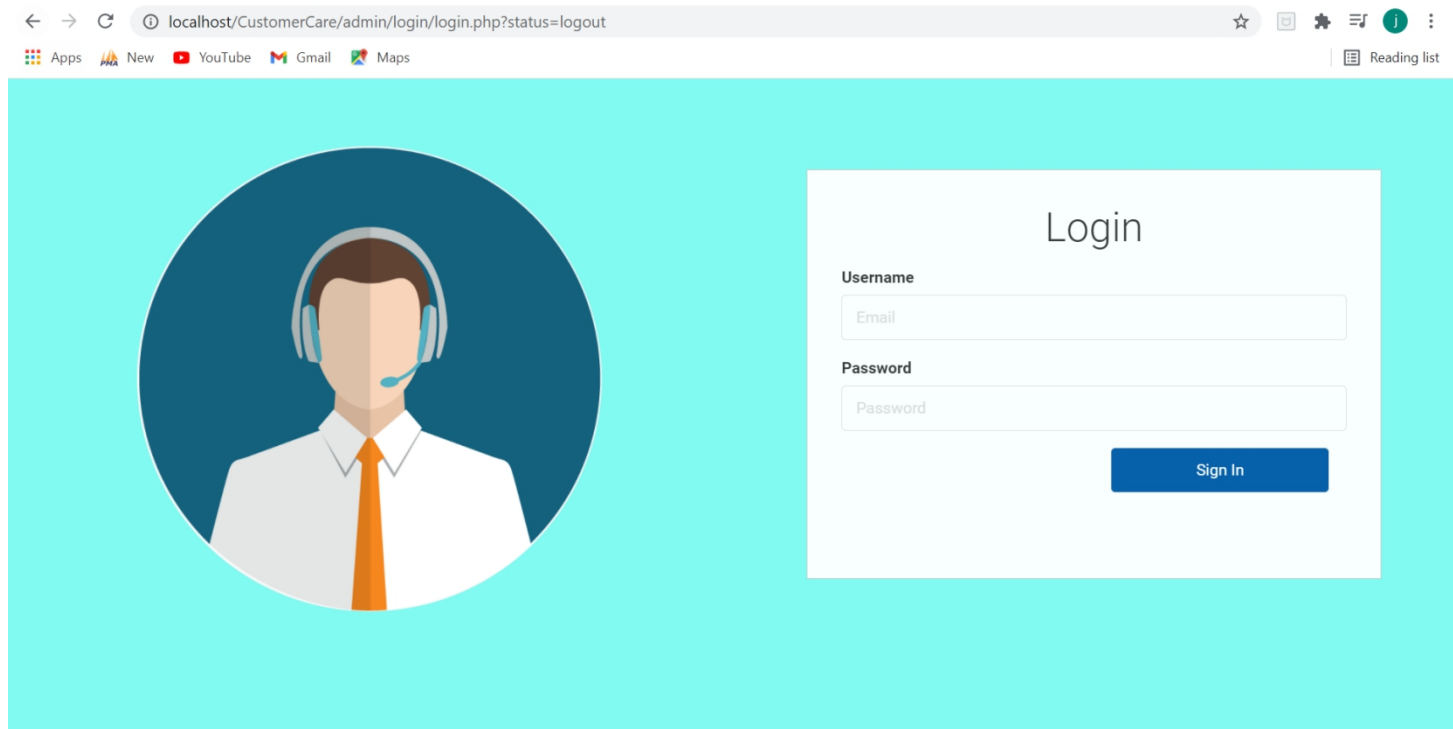


7:26
15
27%

Update Details

Name : testing
Phone : 9467846494
Address : testing
Product : fridge
Customer Comment : testing
Product Comment : testing
Date : 2021-05-03
Status : Follow Up

SUBMIT



localhost/CustomerCare/admin/tbl_customers/select.php

Apps New YouTube Gmail Maps

Reading list



HOME

LOG OUT

Add New

SHOW 10 ENTRIES

SEARCH:

ID	NAME	PHONE	ADDRESS	PRODUCT	CUSTOMER COMMENT	PRODUCT COMMENT	C DATE	C STATUS	EMP ID	UPDATE
1	test	5642586982	test	tedt	tedt	yedtte	2021-04-30	Completed	1	Update
4	ii	666666	hhv bhi	CNN go	CNN	chhi	2021-05-26	Completed	1	Update
5	testing	9467846494	testing	fridge	testing	testing	2021-05-03	Follow Up	1	Update
6	t	9457846134	t	t	t	t	2021-05-05	Follow Up	1	Update
7	ajay	9447508721	test	fridge	test	test	2021-05-05	Follow Up	1	Update
8	gopika	9457816138	test	fridge	test	test	2021-05-08	Follow Up	1	Update
9	farhana	9447508721	test	washing	test	test	2021-05-08	Follow Up	1	Update

localhost/CustomerCare/admin/tbl_customers/form.php

Apps New YouTube Gmail Maps

Reading list



HOME

LOG OUT

Customers Details

ID:

NAME:

PHONE:

ADDRESS:

PRODUCT:

CUSTOMER COMMENT:

PRODUCT COMMENT:

C DATE:

C STATUS:

EMP ID:

[Submit](#) [View All](#)

localhost/CustomerCare/admin/tbl_employee/select.php

Apps New YouTube Gmail Maps Reading list



HOME

LOG OUT

SHOW 10 ENTRIES

SEARCH:

Add New

ID	NAME	PHONE	ADDRESS	EMAIL	DATE OF JOIN	EMP ID	USERNAME	PASSWORD	IMAGE
1	test	9847039693	test	test1234@gmail.com	2-4-2019	e001	test	test	profile.png
2	farhana	9447508721	al harmont , t square road , pachalam	farhana.arehim@gmail.com	26-12-2019	2	farah	qwerty	test1.jpg
3	emp1	09447508721	al harmony, t square road , pachalam	farhana.arehim@gmail.com	12-04-2021	e003	emp	emp	2021-05-09-11-04c1.jpg
4	naseedha	8273738292	test	neenu.shamna@gmail.com	12-04-2021	e004	neenu	neenu	2021-05-12-00-45c1.jpg

Showing 1 to 4 of 4 entries

Previous 1 Next

localhost/CustomerCare/admin/tbl_employee/form.php

Apps New YouTube Gmail Maps Reading list



HOME

LOG OUT

Employee Details

NAME

PHONE

ADDRESS

EMAIL

DATE OF JOIN

EMP ID

USERNAME

PASSWORD

IMAGE

Choose file No file chosen

Submit View All

localhost/CustomerCare/admin/admin_notification/select.php

Apps New YouTube Gmail Maps

Reading list

Home Logout

SHOW 10 ENTRIES

SEARCH:

Add New

ID	ADMIN MESSAGE	MESSAGE DATE	UPDATE	DEL
1	07-Jul-2012 — Select count(distinct(col1 col2)) from mytable where code = a. code... This code uses distinct on 2 parameters and provides count of number of rows specific to those distinct values row count. It worked for me in MySQL like a charm. 19 answers · Top answer: If you are trying to improve performance, you could try creating a persis	2021-04-26	Update	Del
2	test message	2021-05-03	Update	Del
3	test test	2021-05-03	Update	Del
4	100 calls per employee	2021-05-05	Update	Del
5	target 20 customers	2021-05-08	Update	Del
6	call 20 customers today	2021-05-12	Update	Del
7	test	2021-05-13	Update	Del
8	t	2021-05-10	Update	Del
9	call 30 customers today	2021-05-12	Update	Del
10	target 50 calls	2021-05-12	Update	Del

phpMyAdmin

Recent Favorites

There are no favorite tables.

- New
- api
- callcenter
 - New
 - customer
 - emp
- customer_care
 - New
 - admin_notification
 - tbl_customers
 - tbl_employee
- information_schema
- mysql
- performance_schema
- phpmyadmin
- test

Server: 127.0.0.1 » Database: customer_care » Table: admin_notification

Browse Structure SQL Search Insert Export Import Privileges Operations More

Showing rows 0 - 11 (12 total, Query took 0.0058 seconds.)

SELECT * FROM `admin\_notification`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

	id	admin_message	message_date
☐ Edit Copy Delete	1	07-Jul-2012 — Select count(distinct(col1 col2))...	2021-04-26
☐ Edit Copy Delete	2	test message	2021-05-03
☐ Edit Copy Delete	3	test test	2021-05-03
☐ Edit Copy Delete	4	100 calls per employee	2021-05-05
☐ Edit Copy Delete	5	target 20 customers	2021-05-08
☐ Edit Copy Delete	6	call 20 customers today	2021-05-12
☐ Edit Copy Delete	7	test	2021-05-13
☐ Edit Copy Delete	8	t	2021-05-10
☐ Edit Copy Delete	9	call 30 customers today	2021-05-12

Console

Bookmarks Options History Clear

>SELECT * FROM `admin\_notification`

phpMyAdmin

Recent Favorites

- New
- api
- callcenter
- New
- customer
- emp
- customer_care
- New
- admin_notification
- tbl_customers
- tbl_employee
- information_schema
- mysql
- performance_schema
- phpmyadmin
- test

Server: 127.0.0.1 » Database: customer_care » Table: tbl_customers

Browse Structure SQL Search Insert Export Import Privileges Operations More

Showing rows 0 - 27 (28 total, Query took 0.0020 seconds.) [emp_id: 0... - 2...]

SELECT * FROM `tbl\_customers` ORDER BY `emp\_id` ASC

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 50 | Filter rows: Search this table | Sort by key: None

+ Options

			id	name	phone	address	product	customer_comment	product_comment	c_date	c_status	emp_id
☐	Edit	Copy	Delete	12	ww	8596836382	w	fridge	test	test	12-2-2021	follow up
☐	Edit	Copy	Delete	15	fazli	9447508435	test	oven	yt	yt	12-4-2021	not answered
☐	Edit	Copy	Delete	16	renuka	928374692	test	washing machine	w	w	16-5-2021	completed
☐	Edit	Copy	Delete	17	ajay	9867584934	test	OTG	test	yt	2021-03-23	follow up
☐	Edit	Copy	Delete	18	FARHANA	09447508721	al harmony, t square road ,	washing machine	test	yt	2021-05-23	follow up

phpMyAdmin

Recent Favorites

- New
- api
- callcenter
- New
- customer
- emp
- customer_care
- New
- admin_notification
- tbl_customers
- tbl_employee
- information_schema
- mysql
- performance_schema
- phpmyadmin
- test

Server: 127.0.0.1 » Database: customer_care » Table: tbl_employee

Browse Structure SQL Search Insert Export Import Privileges Operations More

Showing rows 0 - 5 (6 total, Query took 0.0011 seconds.)

SELECT * FROM `tbl\_employee`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

				id	name	phone	address	email	date_of_join	emp_id	username	password	image
☐	Edit	Copy	Delete	1	farhana	9847039693	test	test1234@gmail.com	2-4-2019	e001	test	test	profile.png
☐	Edit	Copy	Delete	2	farhana	9447508721	al harmont , t square road , pachalam	farhana.arehim@gmail.com	26-12-2019	2	farah	qwerty	test1.jpg
☐	Edit	Copy	Delete	3	emp1	09447508721	al harmony, t square road , pachalam	farhana.arehim@gmail.com	12-04-2021	e003	emp	emp	2021-05-01-04c1.jp

Console

Bookmarks Options History Clear

>SELECT * FROM `admin\_notification`

CHAPTER 7

7. CONCLUSION AND FUTURE ENHANCEMENTS

7.1 CONCLUSION

- I. “Customer Support” has made it easier for employees to handle their customer calls. The different modules have helped them manage calls very efficiently.
- II. The mobile application helped the employee add new customer details, keep track of customers that needs follow up, update existing customer details based on calls.
- III. The app is designed with an extra module for the manager/admin to add, delete or update employee, customer details and broadcast daily notifications to all the employees.
- IV. Employee have modules to view their details and performance graph.
- V. An extensive set of testing has been done and proved that the application is user friendly and less burden for the employees in an organization.

7.2 FUTURE ENHANCEMENTS

The project has a very vast scope in future. It can be updated in near future as and when requirement for the same arises, as it is very flexible in terms of expansion.

Following are the future scope for the project which would help the employees in an organization to complete their work more efficiently.

1. Call Recording:

The call between the customer and employee can be recorded and stored in database for any future use.

2. Message :

A new module for customer can be added where the customer could login into the app and the employee and customer can communicate through the app.

3. Rate Experience:

Customer could rate their experience with each employee based on which incentive can be given for the employee.

4. Video Call:

During this pandemic time its high risk for an employee to visit a house and provide the customers with services therefore we could provide a module for video call where the employee could help the customers with the appliances.

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